

SAMARTH DIVEKAR

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SUMMARY

Final-year Computer Science & Information Technology (Cybersecurity) student (Sem 7) at SSPU Pune with hands-on experience in **penetration testing**, **vulnerability scanning**, and **AI-powered compliance/security engineering** across three cybersecurity internships. Strong foundation in **Python**, **Node.js/TypeScript** backend systems, **network security**, **CVSS-based vulnerability scoring**, and **OWASP-aligned exploitation testing** (SQLi/XSS). Actively upskilling via **TryHackMe** and **CTF practice**. Interested in threat hunting, adversary TTPs, detection engineering, and AI-driven security tooling.

SKILLS

Programming Languages: Python, JavaScript, Node.js, TypeScript, C, SQL, HTML, CSS

Cybersecurity: Threat Analysis, Penetration Testing, Vulnerability Assessment & Scanning, SQLi/XSS Exploitation Testing, CVSS Scoring, OWASP Top 10 Mapping, Network Security, RBAC Design, Security Auditing, Attack Vector Analysis

Tools: Nmap, Metasploit, Wireshark, Burp Suite, TryHackMe & CTF Practice

Cloud & Infrastructure: AWS (VPC, IAM, Security Groups), Docker, Git/GitHub, REST APIs

Operating Systems: Linux, Windows

AI & Machine Learning: LLM Integration, Prompt Engineering, RAG Systems, Agentic AI Workflows, AI-Powered Compliance Automation

Frameworks & Databases: Flask, Express 5, FastAPI, React, BullMQ, Redis, Prisma, SQLAlchemy, PostgreSQL, SQLite, MongoDB, JSONB

EXPERIENCE

Product Development Lead & Backend Lead | BinQrypt Security

June 2026 - July 2026

- Strengthened DPDPA 2023 compliance readiness by leading backend and database development of **DPDPA InfraMap**, directing **2 database interns** across a **14-table multi-tenant schema** (Node.js 20, Express 5, TypeScript, Prisma 7, PostgreSQL).
- Eliminated a **cross-tenant data isolation vulnerability spanning 6 tables** (missing orgId filters) through systematic investigation and root-cause remediation, preventing cross-organisation data exposure.
- Redesigned access control by migrating **RBAC** from a 3-role rank-based system to a **5-role explicit permission matrix**, and consolidated **3 divergent Prisma schema files** into a single source of truth.
- Enabled secure AI-driven compliance analysis by directing integration of a **self-hosted vLLM instance** on a **private AWS VPC**, with schema-approval gating for auditable data handling.

Penetration Testing Intern | BinQrypt Security

June 2025 - August 2025

- Uncovered exploitable vulnerabilities across web applications and network infrastructure by conducting real-world **penetration testing assessments** in controlled environments.
- Standardized vulnerability triage by scoring all findings against the **CVSS framework**, delivering defensible severity ratings that directly informed **client remediation priorities**.
- Improved knowledge transfer by documenting **attack methodologies, exploited vulnerabilities, and remediation steps** in structured technical reports for client and internal stakeholders.

Cybersecurity & Web Development Intern | Technomind Machine Automation

June 2024 - August 2024

- Strengthened organisational security posture by conducting a comprehensive **security audit** of company web properties, analyzing **common attack vectors** and identifying exploitable vulnerabilities.
- Directly improved remediation outcomes by delivering **actionable, prioritized recommendations** that the organisation adopted to harden its web properties.

PROJECTS

DPDPA InfraMap — AI-Powered Compliance-Gap Analyzer

Backend & Database Lead | Team of 3 | BinQrypt Security

- Directed backend architecture for an **AI-powered DPDPA 2023 compliance-gap analyzer**, designing a **14-table multi-tenant PostgreSQL schema** via **Prisma 7** (Node.js 20, Express 5, TypeScript).
- Eliminated a critical **cross-tenant data leakage vulnerability across 6 tables** by remediating missing tenant-scoping (orgId) filters.
- Redesigned **RBAC** from a 3-role rank-based system to a **5-role explicit permission matrix**, consolidated **3 divergent Prisma schema files** into one, and integrated a **self-hosted vLLM instance** on a **private AWS VPC** with schema-approval gating.

FLOState — Water & Carbon-Aware AI Compute Orchestration

Smart India Hackathon 2026 | Backend & Environmental Scoring Engine Developer

- Designed the **Environmental Scoring Engine**, ingesting live/forecasted **grid carbon intensity, water-stress, and weather data** into a unified **0–100 Compute Cost Score**.
- Built two downstream modules: a **query-complexity classifier** routing simple/complex queries to small/large models, and a **job scheduler** scanning a **48-hour forecast curve** to time-shift batch jobs into low-carbon windows.

- Anchored the solution in the real-world **Google-Adani Visakhapatnam data center water dispute**, motivating the water-stress scoring dimension alongside carbon intensity.
- Built as a **three-layer platform** — LLM query routing, water/carbon-aware scheduling, and a community-impact dashboard — for Smart India Hackathon 2026.

QuantumMail — Post-Quantum Secure Mail Portal

Backend & Security Engineer | Team of 4

- Identified **RSA/ECC email encryption** as vulnerable to quantum computing attacks and engineered a secure mail client as part of a **4-person team**, integrating **Kyber512 (key exchange)**, **Dilithium2 (digital signatures)**, and **AES-GCM (encryption)** via the **Open Quantum Safe (OQS)** library.
- Implemented user authentication, encrypted message transmission, and secure key storage (**SQLite**) through a **CustomTkinter GUI**.
- Built a **phishing-detection component** to analyze suspicious email content and flag malicious indicators.
- Delivered a student-level system using live **NIST-approved post-quantum cryptography standards** in a working, end-to-end product.

VulnScan Pro ("Mini-Nessus") — Modular Web Vulnerability Scanner

Team Member, Scanning & Exploitation Modules | Team of 6

- Contributed to a **13-module client-server vulnerability scanner** (Nessus/OWASP ZAP-inspired), implementing **50-thread port reconnaissance** and **BFS-based form crawling**.
- Built automated **SQLi/XSS exploitation testing** across a **9-step scanning pipeline**, validated against **OWASP Juice Shop and DVWA**.
- Mapped findings to **OWASP Top 10 (2021)/CWE** and generated **ReportLab PDF remediation reports** using a dynamic **0–100 risk-scoring engine**.
- Delivered real-time scan-completion alerts via **Slack webhooks** and **SMTP email notifications** alongside the automated reporting pipeline.

OceanMind AI — Intelligent Ocean Insights Platform

Fusion Engine Developer

- Built the **Fusion Engine**, a multi-source data integration layer unifying siloed oceanic data streams (satellite imagery, IoT sensor streams) for predictive sustainability analysis and marine safety decision-making.
- Placed among the **Top 19 finalists out of 1,000+ teams** by engineering a platform aligned with **UN SDG 13 (Climate Action)**.
- Won the **Special Award and ₹20,000 prize** at the **INDRADHANU International Grand Challenge** (Top 5 finish).
- Enabled real-time monitoring of **ocean stress, illegal fishing activity, and climate-related risk indicators**.

EDUCATION

Bachelor of Technology — Computer Science & IT (Cyber Security)

Symbiosis Skills and Professional University | Pune, India | 2023 – 2027

Sem 5 CGPA: 8.2 | Sem 6 CGPA: 8.5

Higher Secondary Education

Priyadarshani Junior College | Pune, India | 2022 – 2023 | 80%

Secondary Education

Priyadarshani School | Pune, India | 2020 – 2021 | 94%

CERTIFICATIONS

- **Mastercard** — Cybersecurity Job Simulation (Forage)
- **TCS** — Cybersecurity Analyst Job Simulation (Forage)
- **Deloitte Australia** — Cyber Job Simulation (Forage), Enterprise Cyber Incident Investigation

AWARDS & HONORS

- **Special Award & ₹20,000 Prize**, INDRADHANU International Grand Challenge (PCCOE Pune) — Project: OceanMind AI, Top 5 Finish
- **Smart India Hackathon (SIH) 2026** — Ranked 17th out of 200+ Teams, University Level
- **Smart India Hackathon (SIH) 2025** — Top 5 Teams, University Level, advanced to National Selection Round
- **Smart India Hackathon (SIH) 2023** — Second Round Selectee, Top 50 Nationally